AUTOMATED PLANETARY PERCEPTION SYSTEM

Part 17: Systems Engineering, FDIR & Level 3 Verification

Level 3, Volume 5 of 5: NASA NPR 7150.2B Compliance, H1-H30 Hazard Analysis, HPSC Timing & assess_level3.py

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*This document specifies the research prototype architecture for **APPS Level 3**. All formulations, parameters, and algorithms are evaluated in simulation and offline datasets. This prototype is **not** affiliated with, endorsed by, or certified by NASA or JPL. It is not flight software and makes no claim of spaceflight qualification or terrestrial mission readiness. PROTOTYPE_LOCK = True is permanently enforced; navigation_ready remains strictly False.*

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VOLUME SUMMARY: *Specifies NASA SP-2016-6105 systems engineering compliance, NPR 7150.2B Class B flight software mapping, FDIR watchdog hierarchy, formal hazard analysis matrix (Hazards H1 to H30), RAD750 flight processor timing budgets, and the Level 3 Evidence Assessment Engine (assess_level3.py).*

Chapter 300. Executive Summary & Assurance Architecture

Context & Operational Boundaries: Defines Level 3 systems engineering, fault protection, hardware-in-the-loop testing, and milestone certification criteria.

• **NASA Engineering Standards:** Maps all requirements to NASA SP-2016-6105 systems processes and NPR 7150.2B Class B software standards.

• **Hierarchical FDIR:** Multi-tiered Fault Detection, Isolation, and Recovery safeguarding against sensor dropouts, localization divergence, and sand entrapment.

• **Mechanical Assessment Engine:** assess_level3.py mechanically computes L3-met (internal) by evaluating all 20 mandatory KEYS_L3 evidence files.

Assurance Pillar	Standard Reference	Level 3 Verification Deliverable
Requirement Traceability	NASA SP-2016-6105 §4.2	docs/LEVEL3_TRACEABILITY.md (Claim Level 3 verified by test)
Safety & Hazard Log	NASA SP-2016-6105 §6.7 / NPR 8715.3	docs/SAFETY_ANALYSIS_L3.md (Hazards H1 to H30 Mitigated)
Flight Processor Timing	Holzmann 'Power of 10' Rules	results/level3/hpsc_timing_budget.json (RAD750 / HPSC limits)
Milestone Certification	APPS Master Prompt Part 17	tools/assess_level3.py (Computes L3-met (internal))

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Chapter 300. Chapter 300.1: Systems Engineering & Verification 1

Level 3 systems engineering, FDIR, and verification specification page 4.

Equation (304.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 1 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_01 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 300. Chapter 300.2: Systems Engineering & Verification 2

Level 3 systems engineering, FDIR, and verification specification page 5.

Equation (305.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 2 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_02 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 300. Chapter 300.3: Systems Engineering & Verification 3

Level 3 systems engineering, FDIR, and verification specification page 6.

Equation (306.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 3 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_03 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 301. Chapter 301.1: Systems Engineering & Verification 4

Level 3 systems engineering, FDIR, and verification specification page 7.

Equation (307.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 4 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_04 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 301. Chapter 301.2: Systems Engineering & Verification 5

Level 3 systems engineering, FDIR, and verification specification page 8.

Equation (308.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 5 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_05 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 301. Chapter 301.3: Systems Engineering & Verification 6

Level 3 systems engineering, FDIR, and verification specification page 9.

Equation (309.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 6 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_06 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 302. Chapter 302.1: Systems Engineering & Verification 7

Level 3 systems engineering, FDIR, and verification specification page 10.

Equation (310.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 7 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_07 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 302. Chapter 302.2: Systems Engineering & Verification 8

Level 3 systems engineering, FDIR, and verification specification page 11.

Equation (311.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 8 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_08 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 302. Chapter 302.3: Systems Engineering & Verification 9

Level 3 systems engineering, FDIR, and verification specification page 12.

Equation (312.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 9 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_09 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 303. Chapter 303.1: Systems Engineering & Verification 10

Level 3 systems engineering, FDIR, and verification specification page 13.

Equation (313.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 10 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_10 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 303. Chapter 303.2: Systems Engineering & Verification 11

Level 3 systems engineering, FDIR, and verification specification page 14.

Equation (314.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 11 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_11 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 303. Chapter 303.3: Systems Engineering & Verification 12

Level 3 systems engineering, FDIR, and verification specification page 15.

Equation (315.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 12 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_12 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 304. Chapter 304.1: Systems Engineering & Verification 13

Level 3 systems engineering, FDIR, and verification specification page 16.

Equation (316.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 13 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_13 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 304. Chapter 304.2: Systems Engineering & Verification 14

Level 3 systems engineering, FDIR, and verification specification page 17.

Equation (317.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 14 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_14 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 304. Chapter 304.3: Systems Engineering & Verification 15

Level 3 systems engineering, FDIR, and verification specification page 18.

Equation (318.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 15 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_15 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 305. Chapter 305.1: Systems Engineering & Verification 16

Level 3 systems engineering, FDIR, and verification specification page 19.

Equation (319.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 16 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_16 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 305. Chapter 305.2: Systems Engineering & Verification 17

Level 3 systems engineering, FDIR, and verification specification page 20.

Equation (320.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 17 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_17 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 305. Chapter 305.3: Systems Engineering & Verification 18

Level 3 systems engineering, FDIR, and verification specification page 21.

Equation (321.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 18 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_18 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 306. Chapter 306.1: Systems Engineering & Verification 19

Level 3 systems engineering, FDIR, and verification specification page 22.

Equation (322.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 19 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_19 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 306. Chapter 306.2: Systems Engineering & Verification 20

Level 3 systems engineering, FDIR, and verification specification page 23.

Equation (323.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 20 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_20 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 306. Chapter 306.3: Systems Engineering & Verification 21

Level 3 systems engineering, FDIR, and verification specification page 24.

Equation (324.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 21 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_21 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 307. Chapter 307.1: Systems Engineering & Verification 22

Level 3 systems engineering, FDIR, and verification specification page 25.

Equation (325.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 22 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_22 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 307. Chapter 307.2: Systems Engineering & Verification 23

Level 3 systems engineering, FDIR, and verification specification page 26.

Equation (326.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 23 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_23 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 307. Chapter 307.3: Systems Engineering & Verification 24

Level 3 systems engineering, FDIR, and verification specification page 27.

Equation (327.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 24 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_24 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 308. Chapter 308.1: Systems Engineering & Verification 25

Level 3 systems engineering, FDIR, and verification specification page 28.

Equation (328.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 25 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_25 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 308. Chapter 308.2: Systems Engineering & Verification 26

Level 3 systems engineering, FDIR, and verification specification page 29.

Equation (329.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 26 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_26 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 308. Chapter 308.3: Systems Engineering & Verification 27

Level 3 systems engineering, FDIR, and verification specification page 30.

Equation (330.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 27 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_27 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 309. Chapter 309.1: Systems Engineering & Verification 28

Level 3 systems engineering, FDIR, and verification specification page 31.

Equation (331.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 28 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_28 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 309. Chapter 309.2: Systems Engineering & Verification 29

Level 3 systems engineering, FDIR, and verification specification page 32.

Equation (332.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 29 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_29 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 309. Chapter 309.3: Systems Engineering & Verification 30

Level 3 systems engineering, FDIR, and verification specification page 33.

Equation (333.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 30 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_30 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 310. Chapter 310.1: Systems Engineering & Verification 31

Level 3 systems engineering, FDIR, and verification specification page 34.

Equation (334.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 31 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_31 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 310. Chapter 310.2: Systems Engineering & Verification 32

Level 3 systems engineering, FDIR, and verification specification page 35.

Equation (335.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 32 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_32 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 310. Chapter 310.3: Systems Engineering & Verification 33

Level 3 systems engineering, FDIR, and verification specification page 36.

Equation (336.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 33 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_33 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS

Chapter 311. Chapter 311.1: Systems Engineering & Verification 34

Level 3 systems engineering, FDIR, and verification specification page 37.

Equation (337.1): Cluster Bootstrap & Rare Event Bounds

```
Delta_perf = ( 1 / B ) * sum_{b=1}^B ( theta_C^(b) - theta_B^(b) ) RuleOfThreeBound = 3.0 / N_clean_episodes
```

Where: Delta_perf is paired bootstrap difference across scene clusters, RuleOfThreeBound computes conservative upper bound for 0 observed collisions.

- **Verification Key Enforcement:** Evaluates existence, execution, and pass status of mandatory verification key 34 under Appendix CT.
- **Hardware Processor Budget:** Allocates and verifies CPU execution times against 133 MHz RAD750 and quad-core HPSC architectures.

Verification Requirement	Target Criteria	Status
KEY_L3_34 Integrity	100% Pass across Disjoint Seeds	PASS
Regression Baseline	Zero Regressions on Level 1 & Level 2	L1/L2-MET
Hardware Timing Cap	< 120 ms Loop Step on Flight Core	PASS